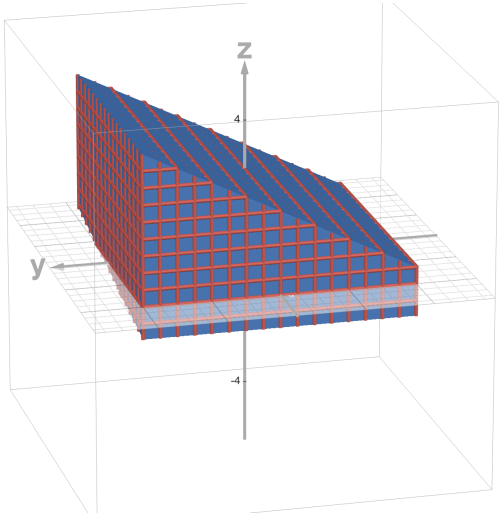
A. **Triple Integrals.** Let E be a solid region.



Suppose the density at position (x, y, z) in the region is

The mass of an infinitesimal bit of volume dV in the region is

We add these inifinitesimal masses up to get the total mass via a **triple integral**.

Or more succinctly:

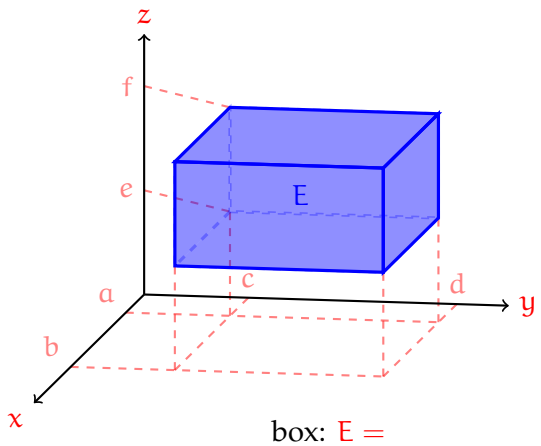
$$\iiint_E [\text{density}]\ dV =$$

In particular, if $[\text{density}] = 1$, then:

and thus

$$\iiint_E 1\ dV =$$

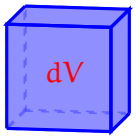
B. **Triple Integrals Over Boxes.** Simple regions of **triple** integration are boxes.



box: $E =$

Let $\delta(x, y, z)$ be the density at position (x, y, z) in box E in mass per volume.

Observe the relationship: $dV =$



First we find the mass in an infinitesimally thin vertical rod through the box.

[mass of vertical rod at fixed x and y] $=$

Next we add up these masses for all x and y in the:

$$[\text{mass of E}] = \iiint_E \delta(x, y, z)\ dV =$$

For a box, we can freely change the order of integration.

For example

$$\iiint_E \delta(x, y, z)\ dV =$$

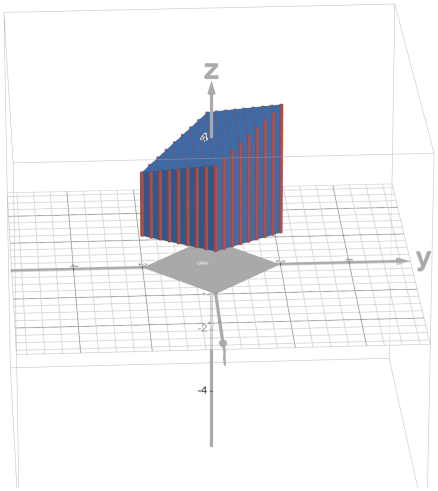
Example 1. If $E = [0,1] \times [0,2] \times [0,3]$ then find

$$\iiint_E 8xyz \, dV$$

Solution. We set this up as an iterated integral in any order.

C. **Triple Integrals Over Other Regions.** Let's consider not boxy regions.

Let E be the region defined by



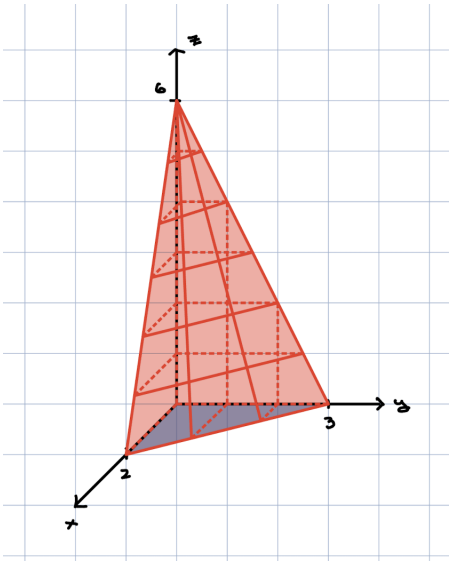
Let $\delta(x,y,z)$ be the density at position (x,y,z) in E in mass per volume.

By first calculating the mass of vertical rods, we have

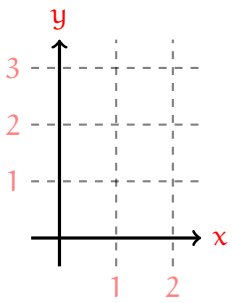
$$[\text{mass of } E] = \iiint_E f(x,y,z) \, dV =$$

Different orders of integration may be possible: instead use horizontal rods.

Example 2. Use a triple integral in order $dzdA$ to find the volume of the **tetrahedron** T bounded by the planes: $3x + 2y + z = 6$, $x = 0$, $y = 0$, and $z = 0$.



Solution. In this order it is first important to identify the xy–shadow.



For each fixed x in:

The range of y in the xy–shadow is:

For each fixed (x, y) in the shadow, the range of z in the tetrahedron is:

$[\text{volume of } E] =$